Assignment 2: Did the Gaza Health Ministry fake casualty numbers?

Start Assignment

- Due Monday by 11:59pm
- Points 20
- Submitting a text entry box or a file upload
- File Types pdf
- Available Sep 25 at 12am - Oct 16 at 11:59pm

This assignment will focus on the topic of casualty counts. To begin, read “How Gaza Health Ministry Fakes Casualty Numbers” (**Wyner, 2024** (https://www.tabletmag.com/sections/news/articles/how-gaza-health-ministry-fakes-casualty-numbers)) by Abraham Wyner (Wharton School, **Department of Statistics and Data Science, Wharton School** (https://statistics.wharton.upenn.edu/profile/ajw/)). Although this is an article written for a more general audience, Wyner is a statistician and expert in sports analytics. In this piece, he argues that the death statistics reported for Gaza are fabricated based on a simple statistical analysis of casualty counts.

All data for this assignment can be accessed **here** (https://github.com/elevien/math50-data/blob/main/notebooks/assignment2_load_data.ipynb) . I do not claim to have vetted these data sources.

Problem 1: Parsing Wyner's Argument

After reading Wyner's article, answer the following questions, pointing to specific lines in the text.

1. What are Wyner's two main observations that lead him to conclude the data are faked?
2. The data in question are counts N_i, M_i, F_i, C_i for the number of casualties of everyone, men, women and children on the ith day. Translate 3 specific observations Wyner makes in his article into probability statements about these variables. Then generate a simulated dataset with these properties in Python and remake the plots in the paper. The data can be real numbers instead of counts. What additional assumptions/modeling choices did you have to make to generate the simulated data?

Problem 2: Regression on Cumulative vs. Daily Deaths

Now consider the following blog post by Lior Pachter (**Pachter, 2024** (https://liorpachter.wordpress.com/2024/03/08/a-note-on-how-the-gaza-ministry-of-health-fakes-casualty-numbers/)), which focuses primarily on the first claim in Wyner's article.

1. In terms of SS_R and SS_{tot}, what is the explanation for a strong correlation with time of a cumulative process?
2. Look at the comment posted by Lior Pachter on March 10, 2024 at 10:51am. What “rule for good science” does this evoke from the Gelman blog post read for the previous assignment? Do you agree that it applies here?
3. What is your reaction to the comment below it, in which someone writes, “Lior the point is not the abundance of anomalies but their inherent contradictions”. Can you think of other explanations for these contradictions?

Problem 3: Examining Real Data

The linked notebook loads Gaza data obtained from the **Tech for Palestine daily casualties API (https://data.techforpalestine.org/docs/casualties-daily/)** into a data frame `gaza`. If you prefer to work with Numpy arrays more like the examples in class, you can use a command like

```
counts = gaza.to_numpy()
```

to obtain numpy arrays with the different columns. It also loads a different data set of Russian losses in Ukraine, compiled from the Ukrainian General Staff's daily reports (**Ivaniuk (https://github.com/PetroIvaniuk/2022-Ukraine-Russia-War-Dataset)**). Of course we expect these two data sets to be very different, but they both contain time series of count data, and it is therefore helpful to ask what statistical patterns may be the same or different and why.

1. Select random sections of this data which have the same number of time points as Wyner used. Does it produce the same result as Wyner's analysis?
2. Let ρ_t denote the correlation coefficient of the daily death counts on a given day and a day t days away. Make a plot of ρ_t as a function of time.
3. Investigate the Ukraine dataset. What are some differences and what are some similarities?

Problem 4: Reflection

In a few paragraphs, reflect on how the analysis of this data may or may not be useful. Does it change your view on the data and whether it is reliable? Consider that the U.S. House of Representatives voted not to cite data from the Gaza Health Ministry. Do you think statistical analysis of the data should be involved in these sorts of discussions? Do you have ideas for additional statistical analysis that would be useful?

References

Department of Statistics and Data Science, Wharton School. Abraham J. Wyner, Faculty Profile. University of Pennsylvania. URL: **https://statistics.wharton.upenn.edu/profile/ajw/ (https://statistics.wharton.upenn.edu/profile/ajw/)** .

Petro Ivaniuk. 2022 Ukraine Russia War Dataset. GitHub. URL: **https://github.com/Petrolvaniuk/2022-Ukraine-Russia-War-Dataset (https://github.com/Petrolvaniuk/2022-Ukraine-Russia-War-Dataset)** . Compiled from the Ukrainian General Staff's daily reports.

Lior Pachter. A Note on How the Gaza Ministry of Health Fakes Casualty Numbers. Bits of DNA (blog), 2024. URL: **https://liorpachter.wordpress.com/2024/03/08/a-note-on-how-the-gaza-ministry-of-health-fakes-casualty-numbers/ (https://liorpachter.wordpress.com/2024/03/08/a-note-on-how-the-gaza-ministry-of-health-fakes-casualty-numbers/)** .

Tech for Palestine. Daily Casualties API. data.techforpalestine.org. URL: **https://data.techforpalestine.org/docs/casualties-daily/ (https://data.techforpalestine.org/docs/casualties-daily/)** .

Abraham Wyner. How Gaza Health Ministry Fakes Casualty Numbers. Tablet Magazine, 2024. URL: **https://www.tabletmag.com/sections/news/articles/how-gaza-health-ministry-fakes-casualty-numbers (https://www.tabletmag.com/sections/news/articles/how-gaza-health-ministry-fakes-casualty-numbers)** .